

The Data Behind the Model: Gaps and Opportunities for Foundation Models in Brain Imaging

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Abstract. Foundation Models (FMs) have revolutionized machine learning in medical imaging, yet their application to brain imaging remains limited and fragmented. Despite the availability of diverse and extensive neuroimaging datasets, most FM research has focused narrowly on a handful of tasks, mainly tumor classification and segmentation, while neglecting prevalent neurological disorders such as ADHD and early-stage Parkinson’s disease. In this work, we present the largest and most comprehensive atlas of brain imaging datasets to date, comprising 151 datasets and over 541k volumetric imaging studies across a wide range of modalities and pathologies. Our meta-analysis of 86 brain imaging FMs reveals a disproportionate reliance on structural MRI and a small set of popular datasets, along with critical blind spots in both disease coverage and imaging modalities. We identify systemic challenges, including inconsistent model evaluation protocols, heterogeneous data formats, and limited availability. All of which hinder reproducibility, scalability, and clinical translation. Our publicly available atlases pave the way for more robust, scalable, and clinically meaningful FMs in brain imaging.

Keywords: Foundation Models · Medical Imaging · Brain diseases.

1 Introduction

Recent advances in machine learning (ML), particularly the development of Foundation Models (FMs), have marked the beginning of a new era in medical imaging. Foundation models are large-scale models trained on large amounts of data, serving as a foundation for various downstream tasks (CLIP, SAM).

Although FMs have recently tackled medical applications, we argue that brain pathologies have been largely ignored in FM research. For example, [8], one of the largest reviews included 200 technical articles, but only 32 covered

brain pathologies; Similarly, the study by Shi et al. [18] provided one of the most comprehensive analyzes and covered 76 research articles, but only ten dealt with brain imaging.

Furthermore, even FMs tackling brain imaging only focus on "popular" ML tasks such as tumor classification or segmentation, missing prevalent neurological disorders like ADHD and early Parkinson predictions. Contrary to the common belief that brain imaging research lacks diverse high-quality datasets and benchmarks to adequately design and evaluate brain FM, we argue on the contrary that (1) a rich variety of brain imaging datasets is already available, (2) current model design and evaluation protocols are not consistent, making fair comparison between FMs difficult, and (3) there is a disconnect between translational applications of medical machine and the main research direction for brain FM.

We support our claims with an evaluation of 151 brain imaging datasets and 86 brain imaging FMs. Our contributions can be summarized as follows.

- We offer the largest brain dataset atlas, compiling imaging datasets totaling more than 541k volumetric studies.
- We provide the largest curation of brain FM, and offer a dataset-oriented meta-analysis.
- We discuss and highlight the main trends and common pitfalls in FM research for brain imaging, and offer a set of recommendations and good practices to start brain imaging FM research.

Given the restrictions on the number of reference pages, we provide complete references of the datasets and models in the atlas.

ATLAS of Brain Image Datasets for Foundation Models (MedAGI)

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Table References

As Name	Ref	Description	Modalities/Sequences	Pathologies	General Pathology	# # Studies	Source URL
ROSMAP	Bennett DA, Buchman AS, E	Religious Orders Study and M	Genomics T2FLAIR T1-MPRAGE DTI SWI (fMRI)	Alzheimer Parkinson Stroke Depression	Vascular pathology Neurod	3414	synapse.org/Sym...
SOOP	Chris Rorden and John Abs	Stroke Outcome Optimization	T1W T2FLAIR DWI	Stroke	Vascular pathology	1715	openneuro.org/...
ATLAS V2.0	Liew SL, Lo BP, Donnelly MI	The entire dataset (N = 1271) is	T1W Segmentation	Stroke	Vascular pathology	955	fcon_1000.project index.html
ISLES22	Hernandez Petzsche, M.R., i	ISLES 2022: A multi-center ma	T2FLAIR DWI T1W T2W Segmentation Demographics	Stroke	Vascular pathology	250	zenodo.org/rec...
FastMRI+	Zhao R, Yaman B, Zhang Y,		T1W T2W T1GdW k-space	Ischemic White Matter Disease Demyelinating Disease Brain Lesions	Vascular pathology Neurod	1001	synapse.org/Sym...
GENETIC REGISTRY	Ph. Muller et al. "GENETIC REGISTRY: Multicenter of HD EEG (2016)		T1W T2W T2GdW	Stroke	Vascular pathology	114	intell.chthonen.n...
COUNT 151						SUM 541012	

Fig. 1: Our interactive atlas is publicly available, and actively updated: <https://ballistic-purple-6f5.notion.site/176bf710ca5980d2bccdd6029523f3ed>

2 Related Work

Foundation Models (FM) are large-scale machine learning models trained on diverse datasets, often unlabeled, using self-supervised learning techniques. They are designed to learn general representations that can be adapted to a wide range

of downstream tasks through fine-tuning. Unlike traditional models built for specific applications, FMs offer scalability, transferability, and efficiency, making them powerful tools for complex domains such as medical imaging.

FMs have gained momentum in medical imaging because of their ability to generalize across tasks with minimal fine-tuning. Pioneering models such as MedCLIP, BioViL, or MONAI’s generative tools have adapted vision language and vision-only pre-training strategies to handle multimodal clinical data. Although these models demonstrate strong performance in tasks such as generating chest radiographs [21], tumor segmentation [17], or improving medical image quality [9,2] most have limited applicability to 3D imaging modalities such as MRI or CT, and few explore domain-specific tasks tailored to neuroimaging.

Only a small fraction of medical FMs explicitly target brain imaging. Existing FM focus primarily on brain tumor segmentation or Alzheimer’s diagnosis. These tasks, while important, represent a limited view of brain-related disorders. Neurological and psychiatric conditions such as ADHD, multiple sclerosis, and Parkinson’s disease are often overlooked. In addition, most studies use limited datasets or lack external validation, reducing generalizability.

To the best of our knowledge, there are no studies on brain datasets for FMs. Curating and analyzing large medical datasets relevant to designing and improving FM for brain research is a critical for FM research, and we believe that our work bridges the gap between datasets publishers, and machine learning researchers, leading the way to more generalizable and reliable brain FM.

3 Methodology

This work aims to provide a comprehensive, structured, and accessible atlas of available brain imaging datasets to support the development and evaluation of brain-specific foundation models. To achieve this, we designed a systematic pipeline to identify, curate, classify, and analyze existing datasets relevant to brain imaging tasks.

3.1 Collection, Identification and Inclusion Criteria

We began by conducting a comprehensive search across major open data repositories (e.g. OpenNeuro, TCIA, Synapse, ADNI, UK Biobank), peer-reviewed literature, and challenge websites. We included datasets that met the following criteria. (1) The primary imaging modality includes the brain (MRI, PET, CT, US, and SPECT. We do not consider EEG signals); and (2) public or institutional access is available with sufficient documentation that includes at least basic metadata (e.g., age, diagnosis, task labels); These criteria led to a curated list of 161 brain imaging datasets.

Next, we surveyed the literature on foundation models by compiling and filtering the models that are relevant to brain imaging from the three largest FM surveys ([8,18]). We curated 86 brain foundation models and analyzed in detail their experimental protocol. In particular, we have collected the datasets

used to train and evaluate each of these models and the tasks for which each model was trained.

3.2 Dataset Annotation Protocol

Each dataset was manually reviewed and annotated with standardized metadata to allow filtering, comparison, and downstream use. We record the following:

- **Name:** Official name of the dataset.
- **Reference:** Citation(s) of original publications or technical reports.
- **Description:** Brief summary of the scope and purpose of the dataset.
- **Modalities/Sequences:** Imaging modalities or sequences included (e.g., MRI, PET, CT, SPECT, US).
- **Specific Pathologies:** Targeted diagnoses or conditions (e.g., Healthy, Parkinson’s, Alzheimer’s, Glioma, Metastasis, Pediatric disorders).
- **General Pathology Category:** Broader classification of diseases (e.g., neurodegeneration, cancer, vascular pathology, mental health condition and other neurological disorders).
- **Studies:** Number of imaging studies (that is, unique imaging sessions per subject, possibly containing multiple series).
- **Source URL:** Link to the official dataset access page.
- **Based On:** If the dataset is derived from or built on another dataset.
- **Anonymization:** Type of anonymization applied (e.g., defaced, skull stripped, deidentified, pseudonymized, fully anonymized).
- **Access Requirements:** Access control level (e.g., public, Data Use Agreement (DUA), Data Transfer Agreement (DTA), restricted).
- **Data Origin Countries:** Country or countries where data was collected.
- **Associated Projects:** Related research projects or consortiums (e.g. ADNI, HCP, UK Biobank).
- **Collection Year(s):** Time period during which data was collected.

Studies, Series, and Sequences In clinical imaging workflows, a study typically refers to a complete imaging session conducted for a patient, which may include several series. Each series comprises a set of image acquisitions performed with a consistent protocol or sequence of imaging parameters.

The fundamental units of a series are slices, individual 2D images that collectively form the dataset. Depending on how the data are structured, the resulting dataset can be organized in different dimensions: 2D datasets contain isolated slices or sections; 3D datasets are composed of volumetric data (stacks of slices); and 4D datasets involve a temporal or contrast-related extension of 3D volumes, capturing variations over time or acquisition parameters.

4 Results

In the following, we present our findings through the percentage and absolute numbers, which total is higher than the total count; it is worth noting that one dataset can cover multiple conditions, multiple modalities, etc. To explore the intersections, we refer to the detailed atlas linked in Sec. 1.

Table 1: Summary of datasets by imaging modality, protocol and tracer. 5

MRI						
Protocol/Contrast	T1	T2	Flair	fMRI	T1Ce	DWI
# Datasets	128	68	60	56	43	26
# Studies	551309	350452	127028	387807	99707	46064
Protocol/Contrast	DTI	DSC	DCE	ASL		
# Datasets	17	10	5	5		
# Studies	65533	77757	743	4270		
PET						
Tracer	AV45	FDG	PIB	Tau	FMISO	Beta DASB
# Datasets	5	4	4	4	1	1
# Studies	5908	3846	4909	7602	423	3368 1100
Tracer	CUMI	Flumazenil	Raclopride	DTBZ	AV133	FE-PE2I
# Datasets	1	1	1	2	1	1
# Studies	1100	1100	298	981	683	100
CT		SPECT		US		
# Datasets	15		2		3	
# Studies	27055		967		22335	

149 *Overview of datasets by modality* Our dataset atlas spans 151 brain datasets en-
150 compassing over 541k volumetric imaging studies, with MRI being the dominant
151 modality both in terms of dataset count and study volume, Tab.1). The most
152 common protocols and contrasts are T1w, T2w, and BOLD/fMRI, while ad-
153 vanced sequences such as DTI, DSC, DCE, and ASL are less frequent. PET spans
154 a wide range of tracers for neurodegeneration (AV45, PIB, Tau), metabolism
155 (FDG, FMISO), psychiatry (DASB, CUMI), epilepsy (Flumazenil) and Parkin-
156 son’s (Raclopride, DTBZ, AV133, FE-PE2I). CT is moderately represented (15
157 datasets) additional SPECT and ultrasound appear in only a few datasets.

158 **In general, the field remains heavily dependent on MRI, with lim-**
159 **ited multimodal coverage (Perfusion) other modalities (PET, CT).**

160 *Overview of Datasets by Medical Condition* A broad spectrum of conditions is
161 covered (Fig. 2). Healthy subjects make up the largest share, with 53 datasets
162 and more than 392,000 studies, forming a strong baseline for control analyzes.
163 Among diseases, Alzheimer’s has the most studies (14 datasets, 38,625 studies),
164 while mental health disorders such as schizophrenia and epilepsy account for 17
165 datasets (34,805 studies). Oncology is also well represented: glioma (27 datasets,
166 12,708 studies), GBM (21, 13,024), metastases (9, 9,503) and other cancers (18,
167 33,458). Stroke (14, 13,443) and Parkinson’s (6, 4,430) add further variety.

168 **Overall, the pathologies’ coverage is very unbalanced. In particular,**
169 **glioma has the most datasets, while Alzheimer’s leads in study volume.**
170 **When aggregated, cancer-related conditions dominate in both dataset**
171 **count and imaging volume. In contrast, neurodegenerative disorders**
172 **(particularly Parkinson’s disease) remain comparatively underrepre-**
173 **sented despite their clinical importance.**

174 *Overview of datasets by impact* We focus in the following on the three patholo-
175 gies with the most datasets, cancer, Alzheimer, and Parkinson diseases (Fig.??).
176 Across all datasets, we notice that the largest datasets (the largest number of
177 studies) are not necessarily the most popular. The largest cancer dataset is
178 NYUMets [12] , while the most cited is the Curious dataset [20] . For both
179 Parkinson and Alzheimer, the relevant dataset with the most citations and stud-
180 ies is the UKBioBank [19] . However, it is worth nothing that this is a commercial
181 dataset and gathers a collection of dozens of cohorts covering many organs and

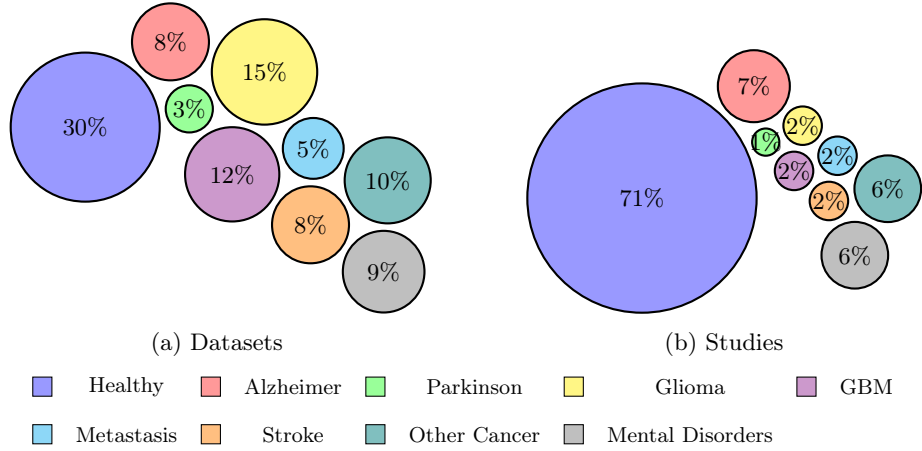


Fig. 2: The distribution of datasets and studies by medical condition

182 pathologies.. For Alzheimer, combining citations and studies in the 4 cohorts
 183 of OASIS [14,11,10,15] leads to a number of studies and citations of magnitude
 184 similar to UKBioBank. It should also be noted that some datasets with a large
 185 number of studies have very few citations (MIRIAD [13] , BrainAge [4] , Rosmap
 186 [1] , or IvyGAP [16]).

187 **There is a mismatch between the citation rate and size of datasets**
 188 **that hints that some important datasets are under-considered.**

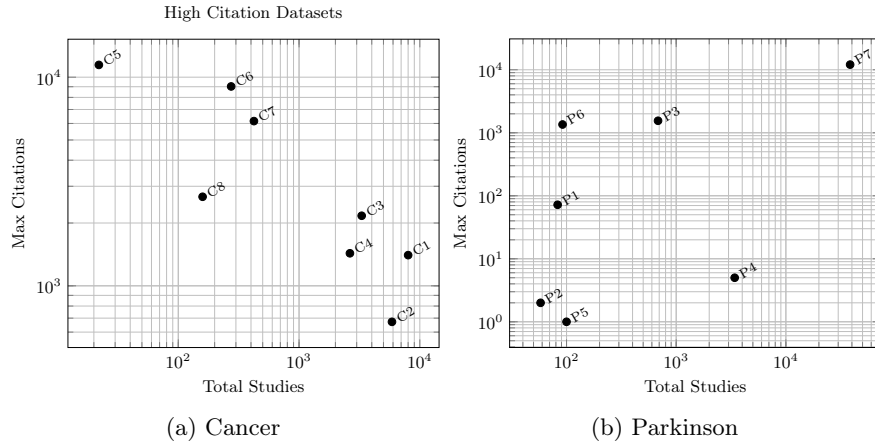


Fig. 3: Discrepancy Between Dataset Size and Impact: Brain imaging datasets for cancer and Parkinson disease, highlighting the four most cited datasets and the four datasets with the largest number of studies for each condition. **Cancer:** C1 NYUMets, C2 BraTS23 Glioma, C3 UPENN-GBM, C4 UCSC Xena Browser, C5 Curious, C6 CGGA, C7 ACRIN-FMISO 6684, C8 LGG-1p19qDeletion. **Parkinson's disease:** P1 INDI-Parkinsons, P2 OpenNeuro-ds004392, P3 PPMI, P4 ROSMAP, P5 OpenNeuro-ds005895, P6 OpenNeuro-ANT, P7 UKBiobank.

5 Discussion

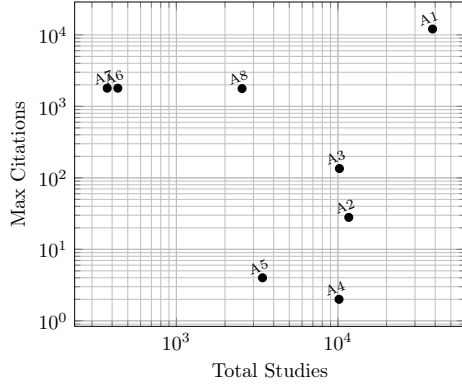


Fig. 4: Discrepancy Between Dataset Size and Impact: Brain imaging datasets for Alzheimer’s disease, highlighting the four most cited datasets and the four datasets with the largest number of studies: A1 UK-Biobank, A2 NACC, A3 iSTAGING, A4 BrainAge, A5 ROSMAP, A6 OASIS-1, A7 OASIS-2, A8 CBMFM

analysis suggests a greater number of public studies focused on neurodegenerative conditions like Parkinson’s and Alzheimer’s compared to those concerning cancer.

Furthermore, our analysis highlights the disproportionate focus of current models on a small subset of diagnostic tasks, particularly brain tumor segmentation and Alzheimer’s classification. Although important, this narrow scope overlooks numerous neurological and psychiatric disorders, including Parkinson’s disease, ADHD, stroke subtypes, and rare pediatric conditions. Broader clinical relevance requires both dataset development and benchmarking efforts that expand beyond these well-trodden tasks.

In general, existing FM prioritize only a subset of available medical imaging datasets, are biased toward brain cancer research, and are less reflective of the actual prevalence of brain pathologies or available brain imaging datasets.

Biases Numerous studies have shown that machine learning (ML) models in medical imaging can exhibit biases, often manifesting as disparities in diagnostic performance between demographic or clinical subgroups [5].

Only six foundational models factored in potential biases when constructing their datasets and models. NeuroBERT [3], for example, curated a balanced dataset reflecting various age groups (18 and older), social classes, and ethnicities (40% non-white). In addition, the authors developed age-specific embeddings to identify age-related irregularities. CLIP-T2Med [7] focused on gender bias during its evaluation. The researchers introduced a novel bias estimation metric to consider gender distribution, showing their method significantly reduces potential gender biases.

Although some biases can be mitigated in FM implementation, other biases inherent to the datasets are much more difficult to tackle. For example, there is a large imbalance in the country of origin of the public datasets collected. 76 datasets accounting for

Our study identified the main patterns and limitations of the available brain imaging datasets that could be used for FM. In the following, we discuss and contrast our insights on brain datasets with our findings from the extended study of brain FMs [6]. We provide a detailed list of FMs online <https://ballistic-purple-6f5 notation.site/182bf710ca5980d6b3a4fd3402960f0e>.

Pathologies coverage. The variety of pathologies represented in FMs differs considerably from what we observed by examining public datasets. Cancer is the most widely represented category in FMs, appearing in 34 models. Vascular diseases are represented in 10 models, just like neurodegeneration, while neurodevelopmental disorders appear in five models, and mental health conditions in three models. Conversely, our dataset

almost half of the brain studies (294.160) are from the USA, the two runners-up, far behind, are Netherlands (8 datasets and 129.054 studies) then China (10 datasets for a total of 37.320 studies).

Another disparity affecting FM generalization caused by the dataset is the discrepancy in the acquisition protocol. In more than half of the datasets (104), no information is provided about the machines or field strength (1.5T, 3T...) used in the collection. These acquisition settings can lead to very different image distributions. Similarly, the underutilization of raw k-space data poses a fundamental limitation: without access to pre-reconstruction data, many foundational tasks, such as learning generalizable representations, simulating artifacts, or improving sequence-specific fidelity, remain out of reach for FM researchers. Only 3 datasets provide raw signal data (RealNoiseMRI, FastMRI, and FastMRI+).

Meta-analysis of best models In today’s landscape, pinpointing the most effective architectures and models for specific tasks and datasets is challenging due to the abundance of datasets and baselines evaluated in various papers, coupled with the absence of standardized benchmarks and datasets in certain contexts. For example, in our study, of 40 FMs with anomaly segmentation, *Brain Tumor Segmentation (BraTS)* is the most popular benchmark, with only 15 models evaluating this dataset in their various iterations (BraTS2018, BraTS2019, BraTS2021, etc.). We illustrate the performance of these FMs in Table 2. The references of each models can be found on the atlases given the page constraints on the reference section. MoME [22], based on mixture of experts is the best performing model. This model uses a hierarchical gating network to combine the predictions of experts. In addition, it prevents the degeneration of each expert network and ensures their specialization using a curriculum learning strategy.

Model ↓	Version	DICE
Med3D	2013	55.41
Med-SAM-Brain	2023	61.85
Voco	2019	78.53
MoME+	2023	83.34
Eff-SAM	2018	83.40
PCRLv2	2018	85.60
MMC-Adapt	2021	86.90
LVM-Med	2021	90.40
Med-SA	2021	90.50
Genesis	2019	90.60
MedSegDiff-V2	2021	90.80
BrainSegFounder	2023	91.15
OBJ-SAM	2023	91.90
MoME	2021	92.10
MAE-Seg	Africa24	92.90

Table 2: FMs evaluated on BraTS, with their dataset variants (BraTS19, BraTS21, ...) and reported DICE scores.

Access and interoperability Another critical challenge is dataset interoperability. The heterogeneity in imaging protocols, metadata formats, anonymization policies, and access terms complicates cross-cohort training and evaluation. For example, one of the largest datasets, UKBioBank, offers narrow commercial licenses, while 13 datasets use Creative Commons Non-Commercial licenses. Meanwhile, for FM, 62 of the 86 models are sharing their source code and only four are restricting commercial usage.

In general, without open, standardized schemas and preprocessing conventions, comparisons between models become difficult, and real-world deployment remains limited. Initiatives to harmonize the dataset schemas, such as common data elements, unified labeling protocols, and flexible input pipelines, could accelerate FM development and benchmarking.

284 *Closing words.* We note a misalignment between the data scale and the impact.
 285 Our results show that the most cited datasets are not necessarily the most extensive,
 286 suggesting that accessibility, usability, and clinical relevance may play a larger role in
 287 adoption than volume alone. Promoting best practices around documentation, access,
 288 and openness will be as important as scaling up imaging archives. Taken together,
 289 our work illustrates that neuroimaging is not suffering from data scarcity, but from
 290 data fragmentation. FMs for brain imaging are expected to benefit substantially from
 291 improved dataset curation, broader task coverage, richer metadata, and lower technical
 292 barriers to adoption. Addressing these challenges is essential not only for technical
 293 advancement, but also for realizing the translational potential of FMs in neuroscience
 294 and clinical care.

295 6 Conclusion

296 Foundation Models (FMs) are reshaping the landscape of AI in medicine, but their
 297 application to brain imaging remains fragmented and narrowly focused. In this work,
 298 we present the most comprehensive atlas of brain imaging datasets to date, spanning
 299 diverse modalities, pathologies, and acquisition protocols.
 300 To guide future research, we curated the largest atlas of brain imaging datasets to
 301 151 datasets comprising more than 541k imaging studies — highlighting a rich but
 302 underutilized landscape of neuroimaging data. Our analysis identified a overwhelming
 303 dominance of structural MRI and a strong reliance on a few popular datasets, alongside
 304 critical blind spots that include underrepresented conditions like Parkinson’s disease
 305 and overlooked data modalities such as advanced perfusion imaging or PET tracers.
 306 Yet, beyond quantity, our findings underscore systemic issues that challenge scalabil-
 307 ity and reproducibility. Brain imaging datasets are highly heterogeneous in acquisi-
 308 tion protocols, annotation formats, and metadata structures. This lack of standard-
 309 ization hinders the development of universal preprocessing pipelines and compromises
 310 cross-dataset model comparison, ultimately limiting FM generalizability. Moreover, the
 311 widespread absence of raw k-space MRI data restricts innovation in fundamental areas
 312 like image reconstruction and physics-informed modeling, slowing progress at the core
 313 of medical image analysis. To unlock the full potential of brain-specific FMs, we advo-
 314 cate for: (1) curated benchmarks with consistent protocols, (2) transparent reporting
 315 and documentation standards, (3) broader inclusion of underrepresented diseases and
 316 modalities, and (4) improved access to raw imaging data, particularly k-space.
 317 We believe that by building on these recommendations, the research community can
 318 shift from fragmented efforts to robust, scalable, and clinically meaningful FMs for
 319 brain imaging.

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